

# PREET PATEL

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**Summary:** Work spans scientific simulation, LLM quality evaluation, biostatistics, and technical evidence review. Seeking roles in agentic evaluations, data science, and physics. First-author research with an M.S. in Physics and a Statistics minor.

## EXPERIENCE

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**Mercor** · *AI Evaluations, Generalist (Contractor)* Apr 2026 – Present

- *LLM Evaluation.* Conduct high-volume human-in-the-loop evaluations of LLM outputs (hundreds to date) for reasoning quality, factual consistency, and instruction adherence, under tight turnaround.
- Regularly added to high-volume batches for consistently doing high quality work, quickly.

**U.S. Patent & Trademark Office** · *Patent Examiner, Photonics & Optical Systems* Oct 2024 – Present

- *Evidence-Based Review.* Synthesize claims, drawings, specifications, and technical literature into traceable written findings and defensible determinations under rigorous quality, compliance, and timeliness standards. Wrote over 100 office actions and several allowances leading to novel inventions being granted patentability.

**GlobalLogic Inc. (Google RMI Team)** · *Data Quality Analyst ("Superrater")* Jul 2024 – Oct 2024

- *Model Quality.* Performed structured evaluations of Gemini and Magi outputs against quality and safety guidelines, identifying reasoning failures, edge cases, and data anomalies.

**University of California, Davis** · *Computational Astrophysicist* Mar 2021 – Sep 2023

- *Big Data & Distributed Computing.* Built reproducible Python and HDF5 analysis pipelines for >1 PB FIRE-2 simulation datasets on national supercomputing systems, improving runtime by 50%+ by refining integration methods, removing redundancies, and improving parallelization.
- *Data Analysis.* Cleaned, transformed, joined, and visualized high-dimensional (20+) scientific datasets to isolate statistically significant patterns and test competing physical explanations (MCMC, gradient descent).
- *Research Leadership.* Led analysis from hypothesis through validation, visualization, and first-author publication in *MNRAS*; collaborated across a 300+ person scientific team with shared software.

## EDUCATION

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**University of California, Davis** · *M.S., Physics* 2020 – 2023

**University of Michigan, Ann Arbor** · *B.S., Physics; B.S., Astronomy & Astrophysics; Minor in Statistics* 2015 – 2019

## PUBLICATIONS & RESEARCH

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**Galactic Archaeology** · *Astrophysics / Big Data* · *MNRAS*, 512(4), 5671

First author. Analyzed >1 PB of high-fidelity FIRE galaxy-formation simulations; predicts observable markers of galactic history embedded in stellar elemental composition.

**Hospitalist Burnout Study** · *Biostatistics / Survey Data* · doi:10.1002/jhrm.70029

Co-author and biostatistician for a 43-participant cross-sectional study; computed Copenhagen Burnout Inventory scores and ran Mann–Whitney *U*, Kruskal–Wallis *H*, and ordered-trend analyses.

**Stellar Spectral-Line Modeling & Uncertainty Quantification** · *Python / MCMC* · [code](#)

Built a likelihood-based workflow for noisy spectra, assessed MCMC convergence, and quantified percentile confidence bounds for recovered signal parameters.

## SKILLS

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*Methods* Statistical inference, MCMC, maximum likelihood, nonparametric tests, regression, time-series analysis, feature engineering, exploratory data analysis, uncertainty quantification, data visualization

*Tools* Python, NumPy, Pandas, Polars, SciPy, Matplotlib, SQL, R, C++, HDF5, Git, Linux/Unix, Excel